

REPORT: OS Security Hardening & Best Practices

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Task: Task 2 – Operating System Security Fundamentals

## 1. Introduction

Operating System (OS) hardening is the process of reducing the "attack surface" of a system. This involves securing user accounts, managing file permissions, and configuring network firewalls. This report outlines key security measures for both Linux and Windows environments.

## 2. User Account Management & Privileges

The most fundamental security step is managing who can access the system.

- **Principle of Least Privilege:** Users should only have the access levels necessary to do their job.
  - *Administrator/Root:* Has full control (can install software, change system settings).
  - *Standard User:* Can run applications but cannot modify core system files.
- **Best Practice:** Always use a Standard User account for daily tasks. Only elevate to Administrator (or use sudo in Linux) when absolutely necessary.

## 3. File Permissions (Linux Focus)

In Linux, file security is controlled by permissions. The command `ls -l` reveals these settings.

- **The Commands:**
  - `chmod` (Change Mode): Used to change read, write, and execute permissions.
    - *Example:* `chmod 700 file.txt` (Only the owner can read/write/execute).
  - `chown` (Change Owner): Used to change who owns the file.
- **Why it matters:** If a sensitive file (like `/etc/shadow` which holds passwords) is readable by everyone, any attacker on the system can steal user credentials.

## 4. Network Security: Firewalls

A firewall controls incoming and outgoing network traffic based on security rules.

- **Windows: Windows Defender Firewall.** It blocks unauthorized apps from accessing the internet.
- **Linux: UFW (Uncomplicated Firewall).**
  - *Command:* `sudo ufw enable` (Turns on the firewall).
  - *Command:* `sudo ufw deny 22` (Blocks SSH access if not needed).

## 5. Service Management

Every running application ("service") is a potential entry point for hackers.

- **Attack Surface Reduction:** We must identify and disable unnecessary services.

- **Windows:** Use "Task Manager" or services.msc to stop unused background services (e.g., Remote Registry).
- **Linux:** Use systemctl list-units --type=service to see running services and systemctl stop [service] to disable them.

## 6. The OS Security Checklist (Deliverable)

Below is the standard hardening checklist derived from this task.

| Category    | Action Item                                | Why?                                       |
|-------------|--------------------------------------------|--------------------------------------------|
| Accounts    | Disable "Guest" accounts.                  | Prevents anonymous access.                 |
| Passwords   | Enforce strong passwords (12+ chars).      | Protects against brute-force attacks.      |
| Updates     | Turn on "Automatic Updates".               | Patches known vulnerabilities immediately. |
| Network     | Enable Firewall (UFW / Windows Defender).  | Blocks unsolicited network traffic.        |
| Storage     | Enable Disk Encryption (BitLocker/LUKS).   | Protects data if the device is stolen.     |
| Services    | Disable unused ports/services (e.g., FTP). | Closes doors that attackers could use.     |
| Permissions | Audit file permissions regularly.          | Ensures sensitive files aren't public.     |

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